DALHOUSIE UNIVERSITY

# CSTCC 2026 Ottawa

## When a Few Trees Dominate the Forest: Outlier Analysis on Thermochemical Benchmarks


Kyle R. Bryenton
Post-Doctoral Research Associate
Dalhousie Department of Chemistry


July 15, 2026

# The GMTKN55 Database

Database for general main group thermochemistry, kinetics, and noncovalent interactions

| GMTKN55 | | | | |
|---|---|---|---|---|
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| ALK8 | PA26 | ISOL24 | ADIM6 | WATER27 |
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Five major categories: basic properties and reaction energies for small systems, reaction energies for large systems and isomerisation reactions, barrier heights, intermolecular noncovalent interactions, and intramolecular noncovalent interactions

$$\text{WTMAD-2} = \frac{1}{N_{\text{bench}}} \sum_{i=1}^{N_{\text{bench}}} \overbrace{\underbrace{\frac{N_{\text{syst},i}}{N_{\text{syst,mean}}}}_{\#\text{ of Systems}} \cdot \underbrace{\frac{\overline{|\Delta E|}_{\text{mean}}}{\overline{|\Delta E|}_{i}}}_{\text{Mean Ref. En.}}}^{w_i^{\text{WTMAD-2}}} \cdot \text{MAD}_i$$

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Goerigk, *et al.*, Phys. Chem. Chem. Phys. **19**, 32184. (2017) doi: 10.1039/c7cp04913g

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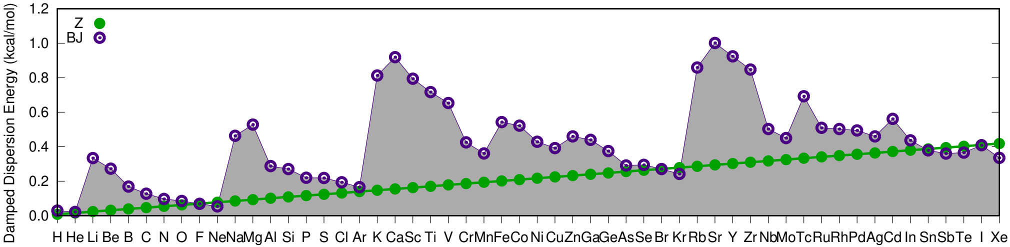
# Axel's One-Parameter Damping Function: Z Damping

To resolve ALK8 issues, Axel replaced the XDM dispersion correction's canonical Becke–Johnson (BJ) damping function in his DH24 double-hybrid functional.

$$E_{\text{disp}}^{\text{XDM}} = -\sum_{i<j} \sum_{n=6,8,10} f_n^{\text{damp}}(R_{ij}) \frac{C_{n,ij}}{R_{ij}^n}$$

$$f_n^{\text{BJ}}(R_{ij}) = \frac{R_{ij}^n}{R_{ij}^n + (a_1 R_{c,ij} + a_2)^n} \qquad f_n^{\text{Z}}(R_{ij}) = \frac{R_{ij}^n}{R_{ij}^n + z_{\text{damp}} \frac{C_{n,ij}}{Z_i + Z_j}}$$




Becke, *J. Chem. Phys.* **160**, 204118. (2024) doi: 10.1063/5.0207682
**Bryenton** & Johnson, *Phys. Chem. Chem. Phys.* **28**, 11626–11638. (2026) doi: 10.1039/D6CP00403B
Rumson, **Bryenton**, & Johnson, *Phys. Chem. Chem. Phys.* **28**, 13543–13552. (2026) doi: 10.1039/D6CP01304J

# Open-Source Implementation and Testing of XDM(Z)

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PAPER

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Cite this: DOI: 10.1039/d6cp00403b

## Consistent GMTKN55 and molecular-crystal accuracy using minimally empirical DFT with XDM(Z) dispersion

Kyle R. Bryenton [ab] and Erin R. Johnson *[abc]

**Bryenton** & Johnson, *Phys. Chem. Chem. Phys.* **28**, 11626–11638. (2026) doi: 10.1039/D6CP00403B
**Bryenton** & Johnson, *Phys. Chem. Chem. Phys.* **28**, 1463–1469. (2026) doi: 10.1039/d5cp03741g

# ~~Open-Source Implementation and Testing of XDM(Z)~~
# Side Quest: Figure out the weird thing happening with WTMAD-2

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Cite this: *Phys. Chem. Chem. Phys.*, 2026, **28**, 1463

## WTMAD-4: a fair weighting scheme for GMTKN55

Kyle R. Bryenton[ab] and Erin R. Johnson*[abc]

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**Bryenton** & Johnson, *Phys. Chem. Chem. Phys.* **28**, 11626–11638. (2026) doi: 10.1039/D6CP00403B
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Introduction

What's the deal with WTMAD-2?!

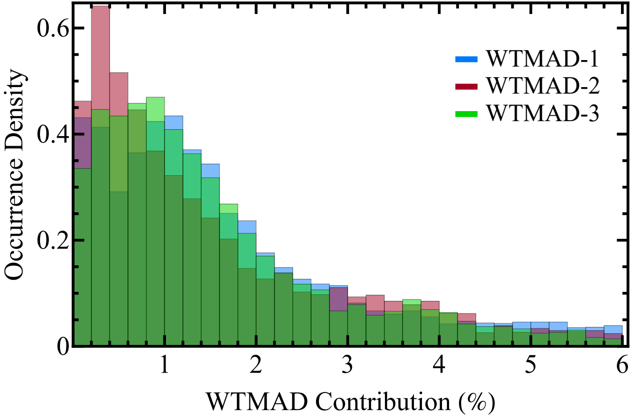
- The issue with WTMAD-2
- Definition of WTMAD-4
- Improvements in Statistics seen with WTMAD-4

Outlier Analysis on Top-Ranked GMTKN55 Functionals

Conclusions

# WTMAD-2 Woes

- Statistics don't form a Gaussian distribution, and have a long decaying tail
- The top 3 benchmarks contribute as much to WTMAD-2 as the bottom 36
- Benchmarks are weighed, on average, $200\times$ greater than others
- Consistent underweighting of some benchmarks (IL16: 0.06%) and overweighting of others (BH76: 9.89%)
- Poses issues for ML or parameter optimisation based on WTMAD-2



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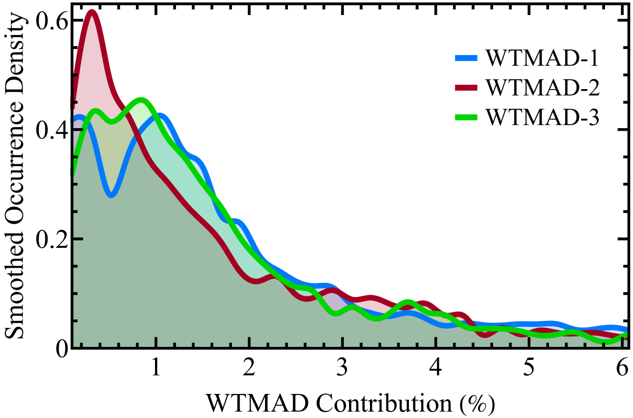


# WTMAD-2 vs WTMAD-4

- These issues are caused by WTMAD-2's dependence on the mean reaction reference energies
- Some benchmarks have $\overline{|\Delta E|}_i \approx 100$ kcal/mol with *small* average errors
- Other benchmarks have $\overline{|\Delta E|}_i \approx 1$ kcal/mol with *huge* average errors

$$\text{WTMAD-}N = \frac{1}{N_{\text{bench}}} \sum_{i=1}^{N_{\text{bench}}} w_i^{\text{WTMAD-}N} \cdot \text{MAD}_i$$

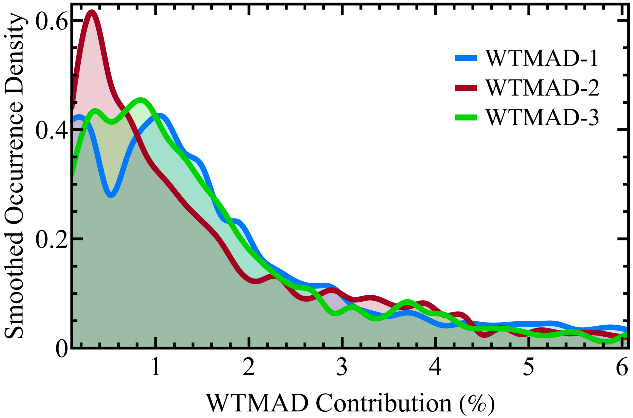
$$w_i^{\text{WTMAD-2}} = \frac{N_{\text{syst},i}}{N_{\text{syst,mean}}} \frac{\overline{|\Delta E|}_{\text{mean}}}{\overline{|\Delta E|}_i} \qquad w_i^{\text{WTMAD-4}} = \frac{100}{N_{\text{bench}}} \frac{3.5}{\overline{\text{MAD}}_i^{\text{10-DFA}}}$$

**The "10-DFA" List (incl. DFT-D3 Dispersion):**

| | | | | |
|---|---|---|---|---|
| PBE0 | B3LYP | B3P86 | PW1PW91 | HSE06 |
| HISS | BHLYP | B3PW91 | MPW1PW91 | LC-$\omega$hPBE |

# WTMAD-4 Wows

- Statistics form a more Gaussian-like distribution centred at $100\%/55 = 1.8\%$
- The top 3 benchmarks contribute as much to WTMAD-4 as the bottom 6
- Benchmarks are weighed typically within a factor of 3
- WTMAD-4 is consistent across all rungs
- Fairer treatment of benchmarks for ML uses or parameter optimisation



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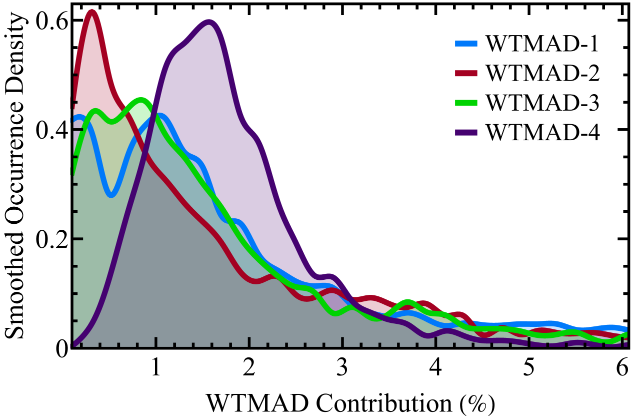

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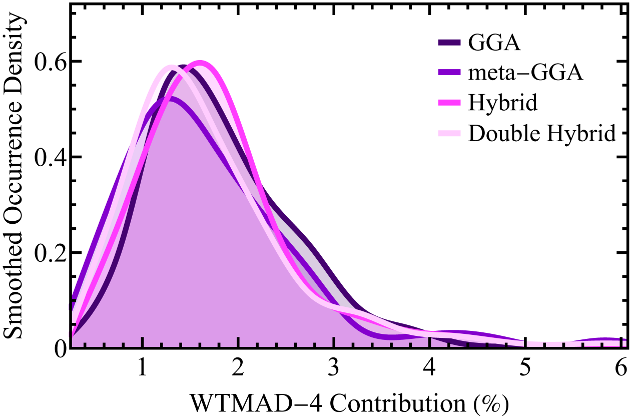

# GGAs and meta-GGA Functionals

## Definitions:

$N_{r>2}$: Number of benchmarks with an error $> 2\times$ *ratio* compared to the "10-DFA" mean

$N_{d>2}$: Number of benchmarks with an error $> 2$ kcal/mol *difference* compared to the "10-DFA" mean

| Functional | WTMAD-$N$ 2 | WTMAD-$N$ 4 | $N_{r\leq 1}$ | $N_{r>2}$ | MAX $r$ | Set | $N_{d>2}$ | MAX $d$ | Set |
|---|---|---|---|---|---|---|---|---|---|
| revPBE-D3(BJ) | 8.24 | 7.67 | 16 | 2 | 2.22 | BHPERI | 9 | 9.70 | SIE4x4 |
| revPBE-XDM(Z) | 8.69 | 7.91 | 19 | 7 | 2.51 | SCONF | 8 | 9.65 | SIE4x4 |
| revPBE-V | 8.50 | 7.94 | 23 | 8 | 2.88 | DIPCS10 | 11 | 9.36 | MB16-43 |
| OLYP-D3(BJ) | 8.76 | 8.16 | 18 | 6 | 2.51 | BHPERI | 13 | 11.95 | SIE4x4 |
| B97-D3(BJ) | 8.49 | 8.32 | 19 | 6 | 2.57 | AL2X6 | 12 | 16.80 | MB16-43 |
| revPBE-XDM(BJ) | 8.40 | 8.45 | 16 | 7 | 3.76 | ACONF | 11 | 9.79 | SIE4x4 |
| B97M-V | 5.46 | 5.24 | 39 | 0 | 1.86 | MB16-43 | 2 | 16.67 | MB16-43 |
| B97M-D4 | 5.68 | 5.44 | 37 | 0 | 1.95 | MB16-43 | 2 | 18.24 | MB16-43 |
| B97M-D3(BJ) | 6.44 | 6.41 | 33 | 3 | 2.81 | AL2X6 | 3 | 4.97 | ALK8 |
| r$^2$SCAN-D3(BJ) | 7.12 | 6.79 | 25 | 0 | 1.72 | WCPT18 | 6 | 4.57 | SIE4x4 |
| B97M-D3(0) | 6.51 | 6.92 | 31 | 3 | 4.39 | ADIM6 | 5 | 28.23 | MB16-43 |
| revTPSS-D3(BJ) | 8.44 | 7.73 | 23 | 3 | 3.71 | SCONF | 9 | 17.45 | MB16-43 |

GGAs: revPBE is still the best GGA functional for molecular chemistry

mGGAs: Most meta-GGAs performed worse than GGAs, and often with much larger max errors

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# Hybrid and Range-Separated Hybrid Functionals

| Functional | WTMAD-$N$ 2 | 4 | $N_{r\leq 1}$ | $N_{r>2}$ | MAX $r$ | Set | $N_{d>2}$ | MAX $d$ | Set |
|---|---|---|---|---|---|---|---|---|---|
| PW6B95-D3(BJ) | 5.50 | 5.14 | 41 | 1 | 2.31 | IL16 | 0 | 1.76 | SIE4x4 |
| M05-2X-D3(0) | 4.62 | 5.19 | 41 | 1 | 2.08 | ADIM6 | 3 | 7.02 | MB16-43 |
| M06-2X-D3(0) | 4.92 | 5.22 | 46 | 2 | 3.50 | HEAVYSB11 | 1 | 5.83 | HEAVYSB11 |
| PW6B95-V | 5.57 | 5.24 | 45 | 2 | 2.49 | ADIM6 | 0 | 1.81 | SIE4x4 |
| revPBE0-D3(BJ) | 5.31 | 5.30 | 43 | 0 | 1.51 | W4-11 | 1 | 3.03 | W4-11 |
| revPBE0-XDM(Z) | 5.71 | 5.43 | 40 | 0 | 1.48 | W4-11 | 1 | 2.83 | W4-11 |
| revPBE0-XDM(BJ) | 5.06 | 5.44 | 42 | 1 | 2.58 | ALK8 | 2 | 7.71 | ALK8 |
| MPW1B95-D3(BJ) | 5.55 | 5.48 | 42 | 1 | 2.02 | ACONF | 0 | 0.85 | RSE43 |
| B86bPBE0-XDM(BJ) | 5.77 | 5.53 | 39 | 0 | 1.93 | ALK8 | 1 | 4.55 | ALK8 |
| M08-HX-D3(0) | 5.27 | 5.68 | 41 | 2 | 3.85 | ACONF | 1 | 2.26 | C60ISO |
| MPW1PW91-D3(BJ) | 6.33 | 5.82 | 32 | 0 | 1.31 | PNICO23 | 0 | 1.04 | ALK8 |
| B86bPBE0-XDM(Z) | 6.43 | 5.85 | 39 | 0 | 1.55 | SCONF | 0 | 1.68 | WATER27 |
| $\omega$B97M-V | 3.18 | 3.72 | 49 | 1 | 2.22 | C60ISO | 1 | 6.51 | C60ISO |
| $\omega$B97M-D4 | 4.02 | 4.34 | 43 | 1 | 2.24 | C60ISO | 2 | 6.60 | C60ISO |
| $\omega$B97X-V | 3.93 | 4.40 | 50 | 1 | 2.57 | C60ISO | 2 | 13.21 | MB16-43 |
| $\omega$B97M-D3(BJ) | 3.94 | 4.60 | 44 | 3 | 2.47 | C60ISO | 2 | 7.86 | C60ISO |
| $\omega$B97X-D4 | 4.33 | 4.68 | 46 | 1 | 2.65 | C60ISO | 1 | 8.83 | C60ISO |

Hybrids: Significant reordering caused by the worsening of mGGA-based hybrids and the improvement of many minimally-empirical ones

RS Hybrids: The $\omega$B97 family performs excellently, but error increased due to large MADs on C60ISO and MB16–43 benchmarks, which WTMAD-2 underweighted

# Hybrid and Range-Separated Hybrid Functionals

| Functional | WTMAD-$N$ 2 | 4 | $N_{r\leq 1}$ | $N_{r>2}$ | MAX $r$ | Set | $N_{d>2}$ | MAX $d$ | Set |
|---|---|---|---|---|---|---|---|---|---|
| PW6B95-D3(BJ) | 5.50 | 5.14 | 41 | 1 | 2.31 | IL16 | 0 | 1.76 | SIE4x4 |
| M05-2X-D3(0) | 4.62 | 5.19 | 41 | 1 | 2.08 | ADIM6 | 3 | 7.02 | MB16-43 |
| M06-2X-D3(0) | 4.92 | 5.22 | 46 | 2 | 3.50 | HEAVYSB11 | 1 | 5.83 | HEAVYSB11 |
| PW6B95-V | 5.57 | 5.24 | 45 | 2 | 2.49 | ADIM6 | 0 | 1.81 | SIE4x4 |
| revPBE0-D3(BJ) | 5.31 | 5.30 | 43 | 0 | 1.51 | W4-11 | 1 | 3.03 | W4-11 |
| revPBE0-XDM(Z) | 5.71 | 5.43 | 40 | 0 | 1.48 | W4-11 | 1 | 2.83 | W4-11 |
| revPBE0-XDM(BJ) | 5.06 | 5.44 | 42 | 1 | 2.58 | ALK8 | 2 | 7.71 | ALK8 |
| MPW1B95-D3(BJ) | 5.55 | 5.48 | 42 | 1 | 2.02 | ACONF | 0 | 0.85 | RSE43 |
| B86bPBE0-XDM(BJ) | 5.77 | 5.53 | 39 | 0 | 1.93 | ALK8 | 1 | 4.55 | ALK8 |
| M08-HX-D3(0) | 5.27 | 5.68 | 41 | 2 | 3.85 | ACONF | 1 | 2.26 | C60ISO |
| MPW1PW91-D3(BJ) | 6.33 | 5.82 | 32 | 0 | 1.31 | PNICO23 | 0 | 1.04 | ALK8 |
| B86bPBE0-XDM(Z) | 6.43 | 5.85 | 39 | 0 | 1.55 | SCONF | 0 | 1.68 | WATER27 |
| $\omega$B97M-V | 3.18 | 3.72 | 49 | 1 | 2.22 | C60ISO | 1 | 6.51 | C60ISO |
| $\omega$B97M-D4 | 4.02 | 4.34 | 43 | 1 | 2.24 | C60ISO | 2 | 6.60 | C60ISO |
| $\omega$B97X-V | 3.93 | 4.40 | 50 | 1 | 2.57 | C60ISO | 2 | 13.21 | MB16-43 |
| $\omega$B97M-D3(BJ) | 3.94 | 4.60 | 44 | 3 | 2.47 | C60ISO | 2 | 7.86 | C60ISO |
| $\omega$B97X-D4 | 4.33 | 4.68 | 46 | 1 | 2.65 | C60ISO | 1 | 8.83 | C60ISO |

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| Functional | WTMAD-$N$ 2 | 4 | $N_{r\leq1}$ | $N_{r>2}$ | MAX $r$ | Set | $N_{d>2}$ | MAX $d$ | Set |
|---|---|---|---|---|---|---|---|---|---|
| PW6B95-D3(BJ) | 5.50 | 5.14 | 41 | 1 | 2.31 | IL16 | 0 | 1.76 | SIE4x4 |
| M05-2X-D3(0) | 4.62 | 5.19 | 41 | 1 | 2.08 | ADIM6 | 3 | 7.02 | MB16-43 |
| M06-2X-D3(0) | 4.92 | 5.22 | 46 | 2 | 3.50 | HEAVYSB11 | 1 | 5.83 | HEAVYSB11 |
| PW6B95-V | 5.57 | 5.24 | 45 | 2 | 2.49 | ADIM6 | 0 | 1.81 | SIE4x4 |
| revPBE0-D3(BJ) | 5.31 | 5.30 | 43 | 0 | 1.51 | W4-11 | 1 | 3.03 | W4-11 |
| revPBE0-XDM(Z) | 5.71 | 5.43 | 40 | 0 | 1.48 | W4-11 | 1 | 2.83 | W4-11 |
| revPBE0-XDM(BJ) | 5.06 | 5.44 | 42 | 1 | 2.58 | ALK8 | 2 | 7.71 | ALK8 |
| MPW1B95-D3(BJ) | 5.55 | 5.48 | 42 | 1 | 2.02 | ACONF | 0 | 0.85 | RSE43 |
| B86bPBE0-XDM(BJ) | 5.77 | 5.53 | 39 | 0 | 1.93 | ALK8 | 1 | 4.55 | ALK8 |
| M08-HX-D3(0) | 5.27 | 5.68 | 41 | 2 | 3.85 | ACONF | 1 | 2.26 | C60ISO |
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| $\omega$B97M-V | 3.18 | 3.72 | 49 | 1 | 2.22 | C60ISO | 1 | 6.51 | C60ISO |
| $\omega$B97M-D4 | 4.02 | 4.34 | 43 | 1 | 2.24 | C60ISO | 2 | 6.60 | C60ISO |
| $\omega$B97X-V | 3.93 | 4.40 | 50 | 1 | 2.57 | C60ISO | 2 | 13.21 | MB16-43 |
| $\omega$B97M-D3(BJ) | 3.94 | 4.60 | 44 | 3 | 2.47 | C60ISO | 2 | 7.86 | C60ISO |
| $\omega$B97X-D4 | 4.33 | 4.68 | 46 | 1 | 2.65 | C60ISO | 1 | 8.83 | C60ISO |

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# Double Hybrids and Machine-Learned Functionals

| Functional | WTMAD-$N$ 2 | 4 | $N_{r\leq 1}$ | $N_{r>2}$ | MAX $r$ | Set | $N_{d>2}$ | MAX $d$ | Set |
|---|---|---|---|---|---|---|---|---|---|
| DH24 | 1.72 | 2.23 | 52 | 0 | 1.57 | C60ISO | 1 | 3.07 | C60ISO |
| revDH23 | 1.72 | 2.26 | 52 | 0 | 1.56 | C60ISO | 1 | 2.99 | C60ISO |
| SOS-DH24 | 1.91 | 2.27 | 52 | 0 | 1.26 | IL16 | 0 | 0.66 | C60ISO |
| SOS-DH23 | 1.95 | 2.34 | 52 | 0 | 1.35 | IL16 | 0 | 0.73 | C60ISO |
| $\omega$DOD-PBEP86-D3(BJ) | 2.20 | 2.50 | 53 | 0 | 1.35 | ADIM6 | 0 | 0.73 | DIPCS10 |
| $\omega$DOD-PBEP86-D3(BJ) | 2.21 | 2.52 | 53 | 0 | 1.54 | ADIM6 | 0 | 0.30 | DIPCS10 |
| DM21mu | 3.93 | 3.72 | 50 | 1 | 2.91 | RG18 | 0 | 1.07 | HEAVYSB11 |
| DM21 | 3.97 | 3.94 | 48 | 2 | 2.68 | RG18 | 1 | 6.01 | C60ISO |
| DM21mc | 3.96 | 3.98 | 49 | 1 | 2.12 | ACONF | 0 | 1.25 | C60ISO |
| Skala | 3.83 | 4.01 | 47 | 0 | 1.60 | C60ISO | 1 | 3.20 | C60ISO |
| DM21m | 3.89 | 4.04 | 48 | 1 | 2.98 | ACONF | 0 | 0.82 | HEAVYSB11 |

D Hybrids: DH24 keeps the top spot, while XYG8 substantially drops from 3rd to 17th due to WTMAD-2 parametrisation

ML DFAs: Perform on par with the best range-separated hybrids, while seemingly having less extreme outliers compared to other highly-parametrised functionals of a similar rung

# Double Hybrids and Machine-Learned Functionals

| Functional | WTMAD-$N$ 2 | 4 | $N_{r\leq 1}$ | $N_{r>2}$ | MAX $r$ | Set | $N_{d>2}$ | MAX $d$ | Set |
|---|---|---|---|---|---|---|---|---|---|
| DH24 | 1.72 | 2.23 | 52 | 0 | 1.57 | C60ISO | 1 | 3.07 | C60ISO |
| revDH23 | 1.72 | 2.26 | 52 | 0 | 1.56 | C60ISO | 1 | 2.99 | C60ISO |
| SOS-DH24 | 1.91 | 2.27 | 52 | 0 | 1.26 | IL16 | 0 | 0.66 | C60ISO |
| SOS-DH23 | 1.95 | 2.34 | 52 | 0 | 1.35 | IL16 | 0 | 0.73 | C60ISO |
| $\omega$DOD-PBEP86-D3(BJ) | 2.20 | 2.50 | 53 | 0 | 1.35 | ADIM6 | 0 | 0.73 | DIPCS10 |
| $\omega$DOD-PBEP86-D3(BJ) | 2.21 | 2.52 | 53 | 0 | 1.54 | ADIM6 | 0 | 0.30 | DIPCS10 |
| DM21mu | 3.93 | 3.72 | 50 | 1 | 2.91 | RG18 | 0 | 1.07 | HEAVYSB11 |
| DM21 | 3.97 | 3.94 | 48 | 2 | 2.68 | RG18 | 1 | 6.01 | C60ISO |
| DM21mc | 3.96 | 3.98 | 49 | 1 | 2.12 | ACONF | 0 | 1.25 | C60ISO |
| Skala | 3.83 | 4.01 | 47 | 0 | 1.60 | C60ISO | 1 | 3.20 | C60ISO |
| DM21m | 3.89 | 4.04 | 48 | 1 | 2.98 | ACONF | 0 | 0.82 | HEAVYSB11 |

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| Functional | WTMAD-$N$ 2 | 4 | $N_{r\leq1}$ | $N_{r>2}$ | MAX $r$ | Set | $N_{d>2}$ | MAX $d$ | Set |
|---|---|---|---|---|---|---|---|---|---|
| DH24 | 1.72 | 2.23 | 52 | 0 | 1.57 | C60ISO | 1 | 3.07 | C60ISO |
| revDH23 | 1.72 | 2.26 | 52 | 0 | 1.56 | C60ISO | 1 | 2.99 | C60ISO |
| SOS-DH24 | 1.91 | 2.27 | 52 | 0 | 1.26 | IL16 | 0 | 0.66 | C60ISO |
| SOS-DH23 | 1.95 | 2.34 | 52 | 0 | 1.35 | IL16 | 0 | 0.73 | C60ISO |
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| $\omega$DOD-PBEP86-D3(BJ) | 2.21 | 2.52 | 53 | 0 | 1.54 | ADIM6 | 0 | 0.30 | DIPCS10 |
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| DM21 | 3.97 | 3.94 | 48 | 2 | 2.68 | RG18 | 1 | 6.01 | C60ISO |
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# Concluding Remarks

1. WTMAD-2 is a misleading statistic of a functional's performance on molecular systems
2. WTMAD-4 is a much fairer representation, but...
3. Boiling 55 benchmarks down to a single number is too simplistic and more careful inspection is essential
4. Minimally-empirical functionals with good physics are still able to perform accurately and consistently

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- Mike Cotnam
- Miko Crouse

ACENET accelerate discovery

Draslovka

# Questions?

## Want My Slides/Thesis?

https://github.com/KyleBryenton/slides-posters

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## WTMAD Boxplots